

Q: What is Materials science and Materials engineering?

Ans:

Materials science is the study of the properties of materials, including their composition, structure, and processing, at the atomic and molecular level. It involves the use of advanced analytical and experimental techniques to understand the behavior of materials and how their properties can be optimized for specific applications.

Materials engineering, on the other hand, is the application of the principles of materials science to design and develop new materials for specific applications. Materials engineers work on developing materials with specific properties, such as strength, stiffness, durability, and electrical conductivity, to meet the needs of various industries, including aerospace, automotive, electronics, and biomedical engineering.

In summary, materials science focuses on understanding the properties of materials, while materials engineering applies that knowledge to develop new materials with specific properties for various applications.

Q: Describe the role of materials in our practical life.

Ans :

Materials science and engineering plays a critical role in our daily lives by providing us with the materials necessary for a wide range of applications.

Some of the key roles that materials play in our practical life are:

1. **Infrastructure:** Materials such as concrete, steel, and asphalt are used in the construction of buildings, bridges, and roads. These materials provide the necessary strength and durability for the infrastructure that supports our daily lives.
2. **Transportation:** Materials such as aluminum, carbon fiber, and advanced composites are used in the construction of vehicles, airplanes, and ships. These materials offer improved performance and fuel efficiency, making transportation more efficient and sustainable.
3. **Electronics:** Materials such as silicon, copper, and gold are used in the construction of electronic devices such as computers, smartphones, and televisions. These materials enable the miniaturization of electronic components and the development of new technologies.
4. **Energy:** Materials such as solar cells, wind turbine blades, and battery electrodes are used in the production and storage of renewable energy. These materials play a critical role in the transition to a more sustainable energy system.
5. **Medical:** Materials such as polymers, ceramics, and metals are used in the construction of medical devices such as implants, prosthetics, and surgical instruments. These materials offer biocompatibility and mechanical properties necessary for medical applications.

Overall, materials science and engineering has a significant impact on our practical lives, providing the materials necessary for a wide range of applications and enabling the development of new technologies that improve our quality of life.

Q: Necessity of Material science & engineering in prospective of Bangladesh.

Ans:

Material science and engineering is a crucial field that deals with the study of the properties, structures, and applications of materials. Bangladesh, being a developing country, also requires the advancements in material science and engineering to boost its economy and ensure sustainable development.

Here are some of the reasons why material science and engineering are essential in the prospective of Bangladesh:

1. **Infrastructure development:** Bangladesh is rapidly growing, and there is a high demand for infrastructure development. Material science and engineering can help in the development of strong, durable, and lightweight materials for the construction of buildings, roads, bridges, and other infrastructure.
2. **Industrial growth:** The manufacturing industry is an essential sector for Bangladesh's economy, and material science and engineering can help in the development of advanced materials for manufacturing processes. These materials can improve efficiency, reduce costs, and increase the durability of products.
3. **Renewable energy:** Bangladesh is exploring renewable energy sources like solar, wind, and hydropower. Material science and engineering can help in the development of advanced materials for energy storage, solar panels, and wind turbines, which can improve the efficiency and reduce the cost of energy production.
4. **Health care:** Material science and engineering can also contribute to the development of new and improved medical devices, implants, and drugs. These advancements can improve the quality of healthcare in Bangladesh and reduce the need for expensive treatments abroad.

5. **Environmental sustainability:** Bangladesh is vulnerable to climate change, and material science and engineering can help in the development of sustainable and eco-friendly materials. These materials can reduce carbon emissions, waste, and pollution, contributing to a better environment for Bangladesh.

In conclusion, material science and engineering can have significant impacts on Bangladesh's economy, industry, healthcare, and environment.

Q : Describe the importance of Material science in modern society.

Ans:

Material science and engineering plays a crucial role in modern society by developing and advancing materials that are used in many different applications. Here are some of the key ways in which material science is important:

1. **Improving existing materials:** Material science is used to improve existing materials to make them stronger, more durable, and more efficient. For example, it has been used to improve the strength and durability of steel, which is used in many different industries.
2. **Developing new materials:** Material science is also used to develop new materials that have unique properties and characteristics. This includes materials such as carbon fiber, which is lightweight and incredibly strong, and graphene, which has excellent electrical and thermal conductivity.
3. **Advancing technology:** Material science is at the heart of many technological advancements, such as semiconductors used in electronics, sensors, and computer chips. It has also played a role in the development of medical implants and prosthetics, as well as solar panels and batteries.
4. **Environmental sustainability:** Material science has an important role to play in creating sustainable materials and processes that reduce waste and pollution. This includes developing biodegradable materials, improving recycling processes, and reducing the environmental impact of manufacturing processes.
5. **Economic impact:** Material science plays a significant role in driving economic growth, as it enables the development of new products and technologies that create jobs and stimulate economic activity.

In summary, material science and engineering are important because they help to improve existing materials, develop new materials with unique properties, advance technology, promote environmental sustainability, and drive economic growth.

Q: Write down the scopes of Material science and engineering.

Ans: Material science is a broad field that deals with the study of the properties, structure, synthesis, and processing of various materials. Some of the significant scopes of material science are:

1. **Construction:** The development of new materials has also led to the creation of stronger and more durable building materials, such as high-strength concrete, steel, and composites.
2. **Nanotechnology:** Material science has led to the development of new materials with unique properties at the nanoscale, which has opened up new opportunities in the field of nanotechnology.
3. **Energy:** The development of new materials has contributed to the creation of energy-efficient technologies, such as solar cells, batteries, and fuel cells.
4. **Aerospace:** The development of lightweight and durable materials has been significant in the aerospace industry, as it has enabled the creation of more efficient and safer aircraft.
5. **Biotechnology:** Material science has led to the development of new materials for biomedical applications, such as artificial organs, drug delivery systems, and tissue engineering.
6. **Electronics:** The development of new materials has led to the creation of new electronic devices and technologies, such as flexible displays, high-performance transistors, and sensors.

In summary, material science and engineering has a vast range of applications and contributes significantly to the development of new technologies and materials, which have a significant impact on our lives.

Q : Why sometime composite are preferable than other materials for engineering application?

Ans: Composite materials are often preferred over other materials for engineering applications because of their unique properties and advantages. Here are some reasons why composite materials are sometimes preferred:

1. **High strength-to-weight ratio:** Composite materials are often stronger and lighter than traditional materials such as metals. This makes them ideal for applications where weight is a critical factor, such as in aerospace, automotive, and sporting goods industries.
2. **Design flexibility:** Composite materials can be designed to have specific mechanical properties and characteristics, such as stiffness, strength, and toughness. This makes them versatile and adaptable to a wide range of applications.
3. **Corrosion resistance:** Composite materials are typically resistant to corrosion and other forms of chemical attack. This makes them ideal for use in harsh environments, such as marine and chemical processing industries.

4. **Thermal insulation:** Composite materials have good thermal insulation properties, which makes them ideal for use in applications that require insulation, such as in the construction industry.
5. **Cost-effectiveness:** Composite materials can often be manufactured using less expensive and more readily available raw materials than traditional materials, which can make them more cost-effective in certain applications.

Overall, composite materials offer unique advantages and properties that make them a preferred choice in many engineering applications.

Q: Discuss the chemical and physical properties of a material.

Ans:

A material's properties can be broadly classified into two categories: chemical properties and physical properties.

Some important chemical properties of materials include:

1. **Corrosion resistance:** This refers to the material's ability to resist chemical degradation due to exposure to corrosive substances such as acids, salts, or bases.
2. **Solubility:** This refers to the material's ability to dissolve in a solvent, which is often used as a means of separating different components of a material.
3. **Reactivity:** This refers to the material's ability to undergo chemical reactions with other substances. For example, some materials are highly reactive and will react with air or water, while others are relatively stable and do not react easily.

Some important physical properties of materials include:

1. **Thermal conductivity:** This refers to the ability of a material to conduct heat, which is important for applications such as thermal insulation and heat transfer.
2. **Electrical conductivity:** This refers to the ability of a material to conduct electricity, which is important for applications such as electrical wiring and electronic devices.
3. **Melting point and boiling point:** These are the temperatures at which a material changes from a solid to a liquid, and from a liquid to a gas, respectively.
4. **Density:** This refers to the mass of a material per unit volume. It is a measure of how closely packed the atoms or molecules are within the material.

In summary, the chemical and physical properties of a material are important factors to consider when designing and selecting materials for various applications in materials science and engineering.

Q: What factors influence the properties of materials?

Ans:

There are many factors that can influence the properties of materials. Some of the key factors include:

1. **Atomic and Molecular Structure:** The atomic and molecular structure of a material can have a significant impact on its properties, such as its strength, hardness, ductility, and conductivity. The arrangement of atoms or molecules within a material can determine the interatomic forces that hold the material together and dictate its mechanical and electrical behavior.
2. **Size and Shape:** The size and shape of a material can also impact its properties, particularly at the nanoscale. For example, nanoparticles may exhibit different chemical and physical properties than their bulk counterparts.
3. **Defects and Imperfections:** Defects and imperfections within a material, such as voids, cracks, and dislocations, can also influence its properties. These defects can affect a material's strength, toughness, and ductility, among other properties.
4. **Composition:** The chemical composition of a material can greatly affect its properties. Different elements and compounds can be added to a base material to alter its properties, such as increasing its strength, durability, or conductivity.
5. **Microstructure:** The microstructure can greatly affect a material's properties, such as its strength, ductility, and corrosion resistance.

Q: Write the basic factors for classification of materials in materials science and engineering.

Ans:

The most basic factors for classification of materials include:

1. **Atomic Structure:** Materials can be classified based on their atomic structure as either crystalline or amorphous. Crystalline materials have a regular, repeating arrangement of atoms or molecules, while amorphous materials lack a long-range order.

2. **Chemical Composition:** Materials can be classified based on their chemical composition, such as metals, ceramics, polymers, and composites.
3. **Mechanical Properties:** Materials can be classified based on their mechanical properties, such as strength, toughness, hardness, elasticity, and ductility.
4. **Electrical Properties:** Materials can be classified based on their electrical properties, such as conductivity, resistivity, dielectric constant, and magnetic properties
5. **Thermal Properties:** Materials can be classified based on their thermal properties, such as thermal conductivity, specific heat, and coefficient of thermal expansion.

These factors are often used in combination to classify materials and identify their properties.

Q: Classify materials according to their properties and uses.

Q: What are the main classes of Engineering Materials.

Ans:

Materials can be classified according to their properties and uses as follows:

1. **Metals:** Metals are characterized by their high strength, ductility, and conductivity. They can be shaped and formed easily and are often used in structural applications, such as construction, automotive and aerospace industries. Metals are also commonly used in electrical applications due to their conductivity, such as wiring and electrical contacts.
2. **Ceramics:** Ceramics are characterized by their high hardness, brittleness, and high temperature resistance. They are used in a wide range of applications, including construction, electronics, and aerospace, due to their ability to withstand high temperatures and harsh environments. Ceramics are also used in medical applications, such as dental implants and artificial joints, due to their biocompatibility and durability.
3. **Polymers:** Polymers are characterized by their low density, high flexibility, and ease of processing. They are lightweight, inexpensive, and have good chemical resistance. Polymers are used in a variety of applications, such as packaging, textiles, and consumer products, due to their low cost and ease of manufacturing. They are also used in medical and biomedical applications, such as implants and drug delivery systems.
4. **Composites:** Composites are made up of two or more materials that have different properties, and are combined to create a material with enhanced properties. They are used in aerospace, construction, and sporting goods. Important characteristics of composites include their ability to be tailored to specific applications, their high strength-to-weight ratio, and their excellent corrosion resistance. Composites are used in a variety of applications, such as aerospace, automotive, and sporting equipment. Carbon fiber composites, for example, are used in high-performance applications such as race cars and aircraft due to their high strength and lightweight properties.

Q: Selection criteria of material for engineering application.

Ans:

The following are some of the main criteria used to select materials for engineering applications:

(1) **Component shape:**

The shape and size of a component has great effect on the choice of the processing unit which ultimately effects the choice of the material. To make it more clear, we consider an example, let the best possible production method is selected, under given conditions, it is die casting, obviously, now the choice of the material becomes limited, ie, one can only choose materials with lower melting points, c.g. aluminium, zinc, magnesion and thermoplastics.

(ii) **Dimensional tolerance:**

There are some materials which can be finished to close tolerance while others cannot Obviously, the required dimensional tolerance for finished components will, influence the choice of materials.

(iii) **Mechanical properties:**

To select a suitable material for specific conditions, all mechanical properties, eg. toughness, hardness, strength, etc. guide us

(iv) **Fabrication (Manufacturing) requirements:**

Method of processing of the material also affects the properties of a component, eg forged components can be stronger than the casted components. Different types of working processes may also give different types of fibre structure. However, investment casting can provide precise dimensions at low cost in comparison to machine operations. Note: fabrication requirements are: castability, i.e.. ease in casting a material, weldability- ease in welding the material, machinability-case to machine a material. formability-ease to form a material, hardenability etc.

(v) Service requirements:

Service requirements are:

- 1. dimensional stability.
- 2. strength,
- 3. toughness,
- 4. heat resistance,
- 5. corrosion resistance,
- 6. fatigue and creep resistance,
- 7. electrical and thermal conductivity etc.

Asif Islam
Mse-19

(vi) Cost:

(A) Cost of the material:

In most of the cases, the cost of raw material accounts about 50 % of the finished cost. Obviously, the cost of the material is a major factor which influences the choice of the material or process. We must note that the use of cheaper material will not always reduce the final cost of the component or product. Use of cheaper material may be associated with higher processing cost due to large number of operations to be performed and also more scrap. We can easily see that this sometimes makes the overall cost more than that of expensive raw material in combination with low processing cost due to lesser number of operations and lesser scrap. The type of material affects the detailed aspect of design) and hence the choice of material as well as the process is selected at the early design state e.g. whether the material is to be joined by spot welding, screws or rivetes, must be decided at the design state.

(B) Cost of processing:

In most of the industries, the processing cost (labour cost) and other costs such as overhead costs account for about 50% of the production cost. Overhead cost in automatic industries is much more than the other costs. If one can somehow reduce all such costs, the total production cost will automatically reduce.

(vii) Availability of the material:

We may find that sometimes the availability of the material becomes a governing factor. When the desired material supply is limited, then a costly material which is available in ample quantity may be chosen.

Q: "understanding the relationship between the structural elements of materials and their properties being an evolution in materials engineering" discuss this statement.

Ans:

Materials engineering is a field that involves the study of materials, their properties, and their applications in various industries. Over time, the understanding of the relationship between the structural elements of materials and their properties has evolved, leading to significant advances in the field.

Traditionally, materials were classified based on their physical and chemical properties. However, as scientists gained a deeper understanding of the atomic and molecular structure of materials, they began to realize that the structural elements of materials play a crucial role in determining their properties.

As materials engineers continue to explore the relationship between structural elements and material properties, they are developing new materials with novel properties that have a wide range of applications in various industries. This includes materials that are lightweight, strong, durable, flexible, conductive, and biocompatible.

In conclusion, the understanding of the relationship between the structural elements of materials and their properties has been an evolution in materials engineering, leading to significant advances in the field and the development of new materials with unique properties that can meet various industry needs.

Q: Suppose you have got a chance to work as a material scientist in a metal processing industry. what would be your role there.

Ans:

As a material scientist in a metal processing industry, my role would be to use my knowledge of materials science and engineering to improve the performance, quality, and cost-effectiveness of the metal products produced by the industry. Here are some of the specific responsibilities I would have:

1. **Research and Development:** My primary responsibility would be to conduct research and development activities aimed at developing new alloys and improving the properties of existing ones. This involves designing experiments, analyzing data, and interpreting results to make recommendations for improvements.
2. **Quality Control:** I would be responsible for ensuring that the metal products produced by the industry meet the required quality standards. This involves conducting quality control tests on the raw materials and finished products to ensure that they meet the required specifications.
3. **Process Improvement:** I would also be responsible for improving the processing methods used to produce the metal products. This involves identifying areas where the process can be optimized to improve efficiency, reduce costs, and improve product quality.
4. **Collaboration:** I would collaborate with other members of the industry, such as engineers, metallurgists, and production managers, to ensure that the production process runs smoothly and efficiently.

In summary, as a material scientist in a metal processing industry, my role would involve conducting research and development, ensuring product quality, improving processing methods, conducting failure analysis, and collaborating with other members of the industry to ensure that the production process runs smoothly and efficiently.

Q: Define matter and materials.

Ans:

1. **Matter:** Matter refers to anything that has mass and occupies spaces. Matter is made up of atoms, which are the smallest units of matter that retain the chemical properties of an element. Matter exists in three states: solid, liquid, and gas, and can be changed from one state to another through physical or chemical changes.
2. **Materials :** Materials are substances made up of matter that are used to make products, structures, or systems. Materials can be natural or synthetic and can be made from a variety of substances such as metals, ceramics, polymers, composites, and semiconductors.

Q: What are Engineering Materials? How does Engineering Materials differ from ordinary materials?

Ans:

Engineering materials are substances and compounds used in engineering to design, manufacture, and construct products and structures. They are selected based on their mechanical, physical, chemical, and thermal properties, as well as cost, availability, and environmental impact. Common engineering materials include metals, polymers, ceramics, composites, and semiconductors.

Engineering materials are specifically designed and optimized for use in engineering applications. They are chosen based on their unique properties and characteristics that make them suitable for specific applications. Ordinary materials, on the other hand, are materials that are commonly found and used in everyday life, but may not have been specifically designed for engineering purposes. Engineering materials undergo rigorous testing and analysis to ensure that they meet the required standards for their intended use, while ordinary materials may not have undergone such scrutiny. Additionally, engineering materials are often subject to strict regulations and standards, while ordinary materials may not be subject to such regulations.